Soumyashree Sahoo

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Profile

Final-year B.Tech (CSE – ICS) student passionate about Penetration Testing, VAPT, and Red Teaming. Skilled in simulating attacks, exploiting vulnerabilities, and building custom tools for security assessments. Hands-on with real-world CTFs, offensive tooling, and vulnerability reporting. Eager to apply technical expertise to secure modern systems and contribute to high-impact red team engagements.

Skills

Languages: Python, C, HTML, CSS, JavaScript
Tools & Frameworks: Burp Suite, Nmap, Wireshark, Metasploit, Splunk, IDS/IPS
Security Expertise: Penetration Testing, Red Teaming, Web Application Security, Vulnerability Assessment & Management, Network Security, API Security
Platforms: Kali Linux, HackTheBox, TryHackMe, picoCTF

Education

B.Tech in Computer Science & Engineering (ICS) - Sambalpur University Institute of Information Technology (2022 – 2026)

Experience

Project Intern – NIT Puducherry (On-site) | May 2024 – Jul 2024

- **Developed a scalable 360° video analytics pipeline :** Engineered a multi-threaded processing framework that segmented panoramic feeds into six parallel Fields of View (FOVs), enabling concurrent object detection and tracking using YOLOv9 and Kalman filters for improved spatial consistency.
- **Optimized multi-view data synchronization and preprocessing:** Utilized OpenCV and NumPy to enhance frame alignment and data throughput, achieving a 32% reduction in latency and sustaining real-time inference at 24 FPS across multi-camera streams.
- **Benchmarked system performance and documented workflow:** Conducted extensive performance analysis across 10+ datasets, recording an 18% accuracy improvement compared to the baseline model, and prepared documentation outlining scalability and optimization results for research integration.

Projects

ProofVault (Blockchain-based Evidence Verification Platform)

- **ProofVault** ensures tamper-proof digital evidence verification through IPFS-based decentralized storage and SHA-256 hash integrity checks. Implemented a FastAPI microservice for authenticated uploads, on-demand validation, and Solidity smart contracts for on-chain timestamping and chain-of-custody tracking. Delivered a 48% reduction in verification latency and 2× throughput improvement over centralized approaches while maintaining cryptographic auditability.

EntropySense (Randomness Generation Using Road Traffic Datasets)

- **EntropySense** extracts cryptographic entropy from real-world road traffic video datasets sourced from Kaggle and Roboflow. Designed frame-level analysis pipelines leveraging Python, OpenCV, and NumPy to derive 8-bit entropy sequences, powering AES-256 based file encryption in a secure upload platform. Achieved 60% faster encryption upload cycles compared to PRNG-based methods and validated entropy strength with NIST SP 800-22 and Dieharder tests, passing 99% of uniformity benchmarks.

Certifications

- Google Cybersecurity
- Cisco: Introduction to Cybersecurity
- Linux Fundamentals (LearnQuest)
- API Security Fundamentals

Additional Contributions

- Medium Blog: Authored articles on penetration testing, CTF walkthroughs, and tool usage.
- Solved Various CTFs - PicoCTF, TryHackMe, HackTheBox
- Open Source: Hacktoberfest 2024 contributor.